

ADBMS – Assignment 1

Histogram

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1 Database

The database used for the experiment is the Join Order Benchmark (2013 Snapshot). The table cardinalities were obtained using `COUNT(*)`. For this experiment, the primary-key column is recorded as `id` for all tables.

2 Q1 – PostgreSQL Histograms

The PostgreSQL histograms for the `id` and `title` columns of the `title` table were examined using `pg_stats`.

2.1 Q1.1 Corresponding SQL

The following query retrieves the statistics maintained by PostgreSQL:

```
SELECT
    attname ,
    null_frac ,
    n_distinct ,
    correlation ,
    histogram_bounds
FROM pg_stats
WHERE tablename = 'title'
    AND attname IN ('id', 'title');
```

The statistics obtained for the two columns were:

Column	Null fraction	n_distinct	Correlation	Histogram buckets
id	0.0	-1.0	0.99812114	100
title	0.0	139335.0	0.40523243	100

Table 1: PostgreSQL statistics for `title.id` and `title.title`

For `id`, PostgreSQL reported 101 histogram boundary values, corresponding to 100 histogram buckets.

For `title`, PostgreSQL reported 101 boundary values, corresponding to 100 histogram buckets.

2.2 Q1.2 Type of Histogram

PostgreSQL uses an equi-depth style histogram for these statistics. The histogram boundaries divide the ordered sample into approximately equal numbers of observations per bucket.

2.3 Q1.3 Number of Buckets

The number of buckets represented by the PostgreSQL statistics is:

- **id:** 100 buckets
- **title:** 100 buckets

2.4 Q1.4 Representation of the Textual Attribute

The **title** attribute is textual, but PostgreSQL can still order text values according to the database's comparison rules. Therefore, the values can be sorted and histogram boundaries can be selected from that ordering.

The histogram does not convert the title into a numerical value for the purpose of the histogram. Instead, the boundary values themselves are strings, and PostgreSQL uses its text comparison semantics when estimating predicates involving the column.

2.5 Q1.5 Histogram Bucket Boundaries

ID boundaries

The PostgreSQL **id** histogram contains 101 boundary values:

```
{114,24573,50079,74448,102250,129121,155133,181016,206149,
233863,260767,287742,314506,339108,362177,386919,411783,
435703,459964,485728,512026,535879,560447,587620,614036,
640736,663165,686493,710381,733376,759225,782723,806012,
829052,852028,877501,903058,927725,953127,978610,1004595,
1032084,1058076,1083746,1108234,1131865,1155911,1179238,
1205508,1234082,1259069,1284674,1310759,1335478,1360463,
1386675,1412692,1435435,1462459,1488257,1514089,1541398,
1568648,1593315,1617722,1642842,1665716,1690730,1717894,
1739710,1765639,1790886,1817383,1842017,1864212,1890074,
1915985,1943166,1967914,1997705,2023158,2051349,2076415,
2103407,2127836,2151710,2175304,2201304,2224706,2251068,
2274191,2297960,2324975,2350972,2378017,2401656,2427498,
2477404,2502294,2528164}
```

Title boundaries

The PostgreSQL **title** histogram contains 101 boundary values. The boundary values are strings ordered according to PostgreSQL's text comparison rules.

```
{"'Allo 'Allo!",(#1.140),(#1.254),(#1.4825),(#1.80),
(#13.62),(#2.36),(#3.70),(#55.74),(#9.6),(1974-03-04),
(1990-01-27),(1996-07-06),(1999-05-28),(2001-01-17),
(2002-08-20),(2004-03-19),(2005-07-16),(2006-11-20),
(2008-02-07),(2009-09-08),(2011-02-26),(2012-03-27),
(2013-02-22),"30 Years of Emmerdale","A Lesson in Murder",
Abgeschminkt,"Akordet e Kosovës - populllore",
"An Autopsy of The Killing","Arrival of Rex",
"Bad Luck Be a Camper Tonight/Step Clam",
"Ben and Burman Find God","Black Bunbusters",
"Boy from Woy","Caffe Lena","Chamana Atha Guntha",
"Cities of Refuge",Cortège,"Dancing Arabs","Deadly Advice",
"Der Umweg",Dinty,"Dr. Hugo","Ein Bett für Drei",
"Empress of Ireland","Extremely Yours, Stryc-9",
"Final Approach","Frau Flinz","Genius of the Jet",
```

"Goode Lovin'", "Hard Times in the Country", Highlights,
 "Hotel Lolita 8", "If Lovin' You Is Wrong, I Don't Wanna Be Rat",
 "Internet Zombies!", "Jed and the Countess", Kadetten,
 "Kirameite, watashi no naka no watashi", "L'ombre d'un doute",
 "Lagda Ishq Ho Gaya",
 "Leila Shems: Everyone Is Invited to the Pants Party Except Me",
 "Little Big Head Man", "Love Thy Neighbor",
 "Männer soll man nicht alleine lassen", "Ménage à Trois",
 "Mob Wives Chicago", "Mud Show", Naujawan, "Nobody's Darling",
 "Oh Brother/Taz Babies", Otryv, Path-o-Prasad, Pivete,
 "Prince William Reveals How the Queen Helped Plan His Wedding",
 "Rafferty's Rise", Reunions, "Royal Cat Nap", "Sati Pingala",
 "Sentenza di morte", "Show Biz",
 "Snow White and the Juniper Tree", "Srdele, lignje i leptiri",
 "Summer Solstice Festival", "Tap Time", "The Annihilators",
 "The Case of the Horrifying Virus: Part 1", "The Erased",
 "The Haunted Planet", "The Living City",
 "The Nomination of Abraham Lincoln", "The Sandman",
 "The Tramp and the Tenner", "This Is My Past",
 "Top 2 Performance", Turnaround,
 "Une cicatrice derrière la tête", Vertauscht,
 "Wave Skiing 2.0 Jaws", "Who Killed Me?", "Y-2 Pappy",
 "Zzaap and the Word Master"}

2.6 Q1.6 Histogram Plots

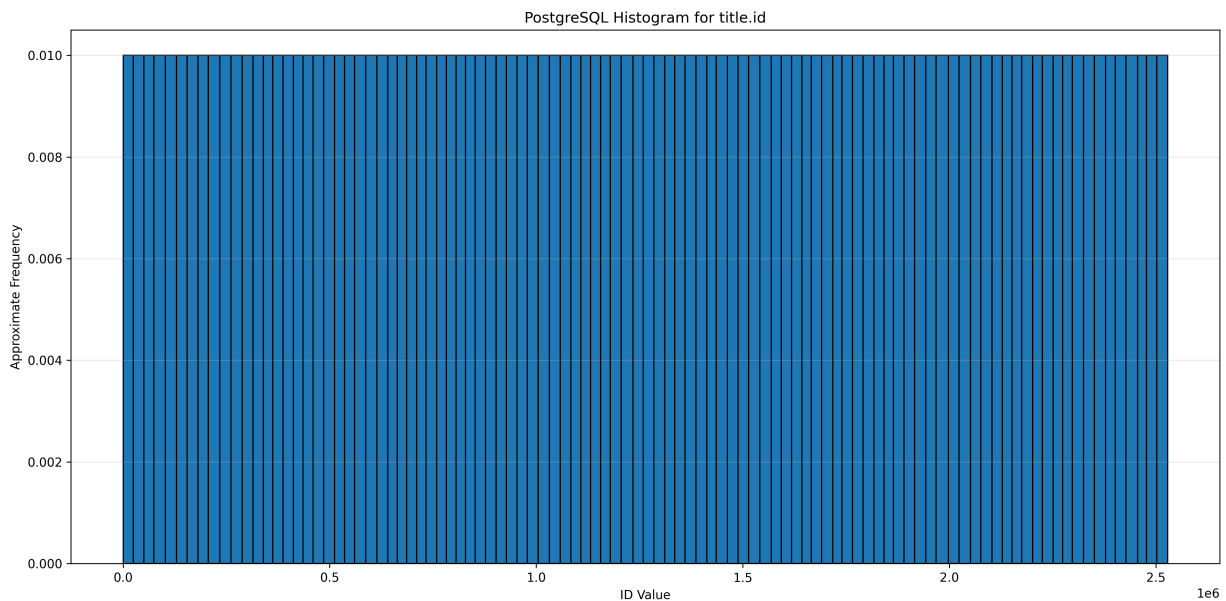
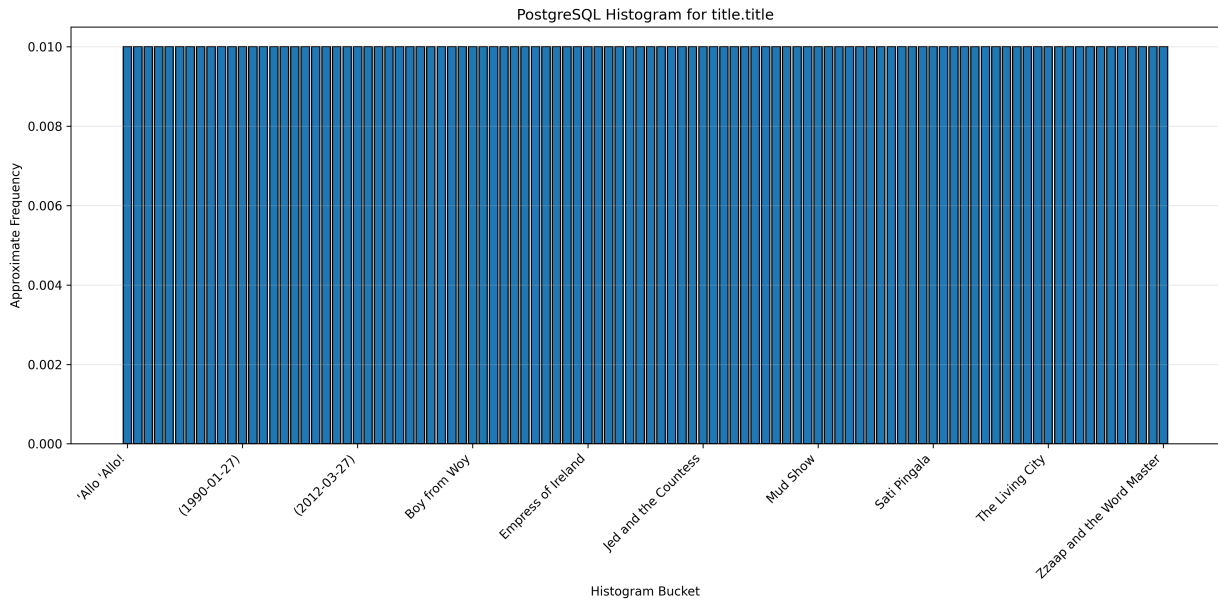


Figure 1: PostgreSQL histogram for title.id

Figure 2: PostgreSQL histogram for `title.title`

2.7 Q1.7 Maximum Selectivity Error

For a query Q , the selectivity error is

$$\text{Selectivity Error}(Q) = |S_{\text{estimated}}(Q) - S_{\text{actual}}(Q)|.$$

The maximum selectivity error is

$$E_{\max} = \max_Q |S_{\text{estimated}}(Q) - S_{\text{actual}}(Q)|.$$

For the Q1 experiment, the maximum selectivity errors obtained were:

$$E_{\max, \text{id}} = 0.004393, \quad E_{\max, \text{title}} = 0.064477$$

2.8 Q1.8 Selectivity Estimation for a Text Range Query

For a predicate such as

```
title < 'Race'
```

PostgreSQL determines the position of the string 'Race' in the ordered histogram. It identifies the bucket containing the value and estimates the fraction of values below the predicate value using the histogram information.

The estimate therefore depends on the histogram boundaries and the relative position of the query value within the appropriate bucket.

2.9 Q1.9 Estimated and Actual Selectivities

Query 1

```
SELECT *
FROM title
WHERE title.title < 'Race';
```

Estimated selectivity:

$$S_{\text{estimated}} = 0.758368$$

Actual selectivity:

$$S_{\text{actual}} = 0.759357$$

Therefore,

$$E = |0.758368 - 0.759357| = 0.000989.$$

Query 2

```
SELECT *  
FROM title  
WHERE title.id < 350000;
```

Estimated selectivity:

$$S_{\text{estimated}} = 0.134734$$

Actual selectivity:

$$S_{\text{actual}} = 0.138432$$

Therefore,

$$E = |0.134734 - 0.138432| = 0.003698.$$

3 Q2 – Optimal Serial Histogram

A serial optimal (V-optimal / minimum-variance) histogram was implemented using dynamic programming with **20 buckets**. The experiment uses target sizes 1000, 3000, and 5000. For **id**, these targets refer to sampled rows and therefore also to distinct values. For **title**, these targets refer specifically to the number of **distinct titles**.

For **title**, the sampler keeps collecting rows until the required number of distinct titles is reached. Duplicate title rows are retained because their frequencies are required for histogram construction. Thus, the actual raw row count can exceed the distinct-value target.

Each target was evaluated using 3 independent fresh random samples. Build time and maximum selectivity error were averaged over the three trials. Row-count and distinct-value checks were performed before histogram construction.

3.1 Q2.1 Histograms

The following are the **vertical** histogram plots. Each vertical bar represents one of the 20 buckets.

3.1.1 ID – 1000 samples

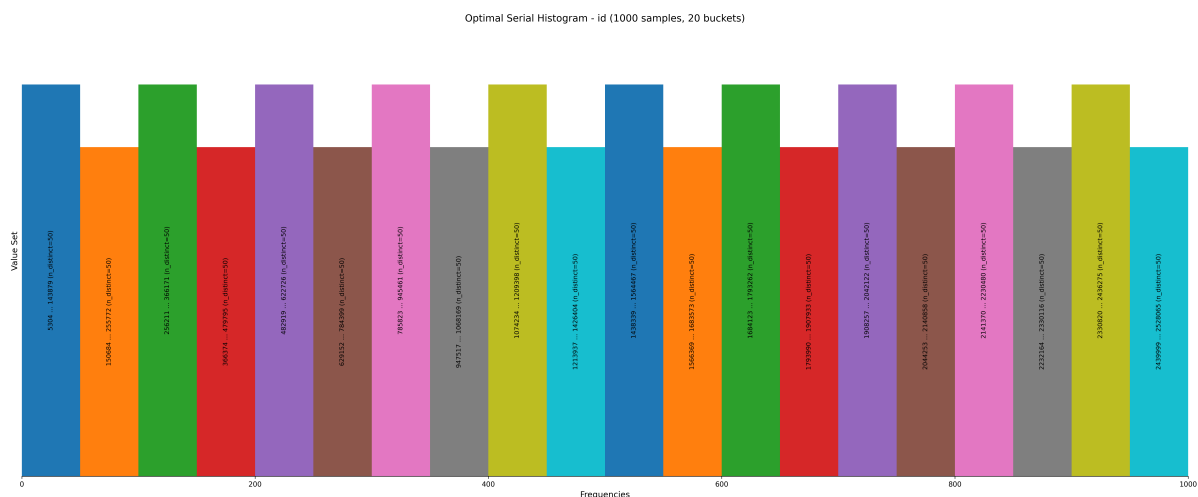


Figure 3: Optimal serial histogram for id, 1000-sample target, 20 buckets

3.1.2 ID – 3000 samples

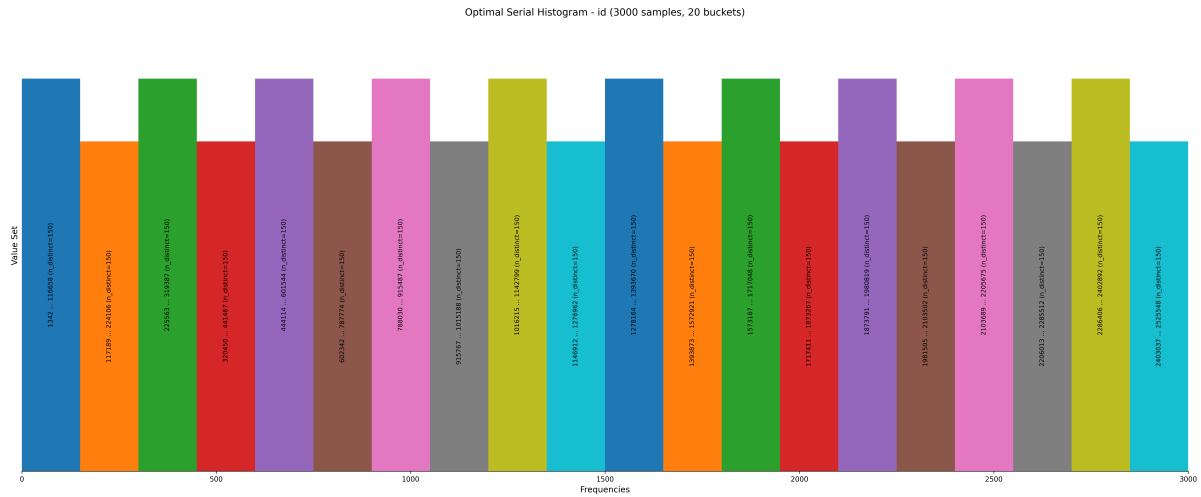


Figure 4: Optimal serial histogram for id, 3000-sample target, 20 buckets

3.1.3 ID – 5000 samples

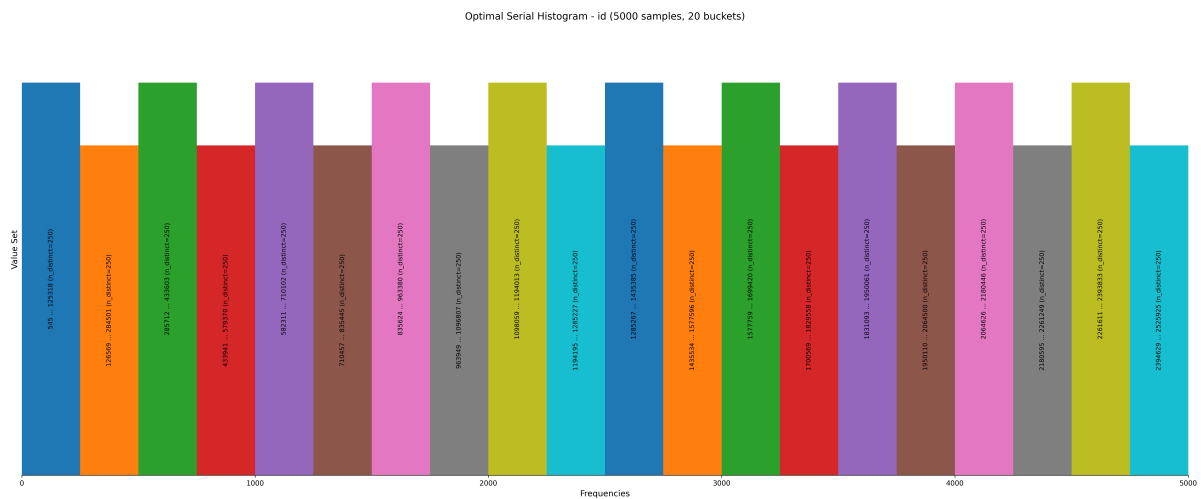


Figure 5: Optimal serial histogram for id, 5000-sample target, 20 buckets

3.1.4 Title – 1000 distinct titles

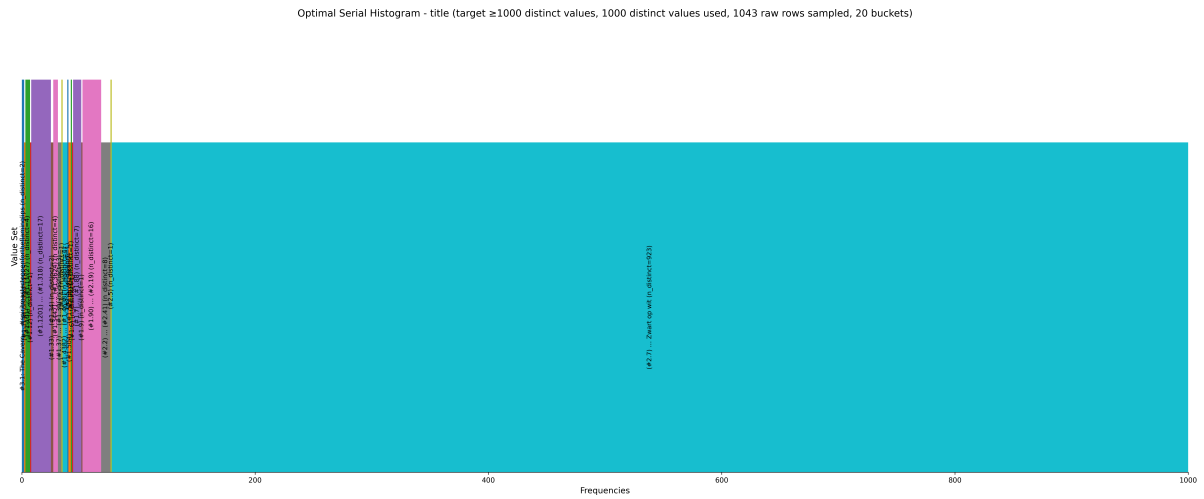


Figure 6: Optimal serial histogram for `title`, 1000 distinct-title target, 20 buckets

3.1.5 Title – 3000 distinct titles

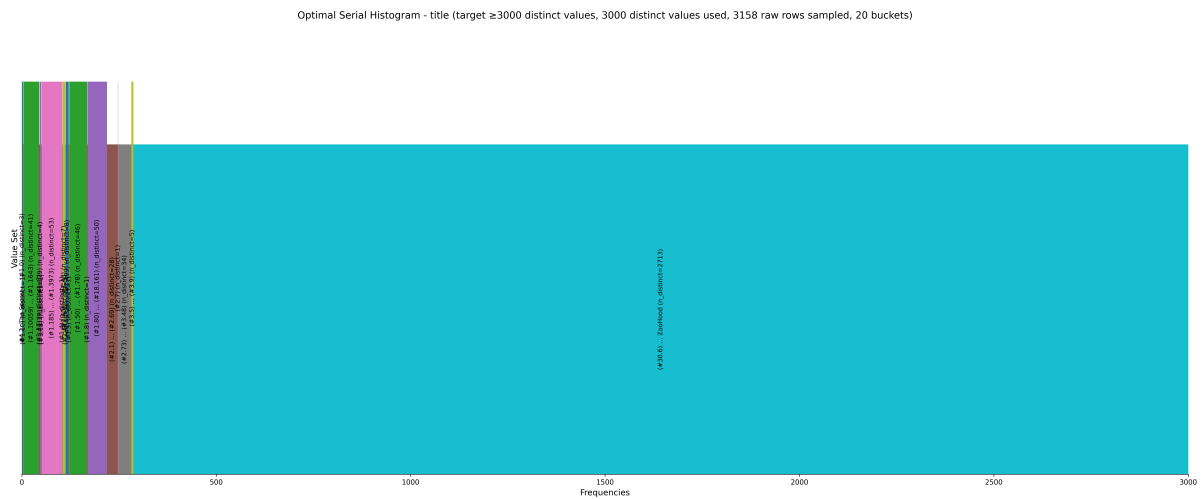


Figure 7: Optimal serial histogram for `title`, 3000 distinct-title target, 20 buckets

3.1.6 Title – 5000 distinct titles

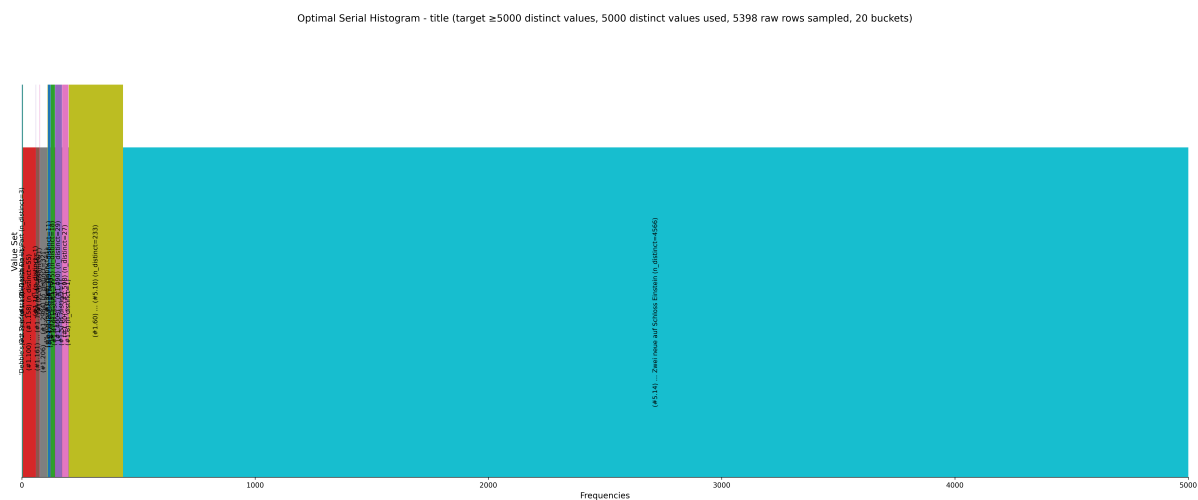


Figure 8: Optimal serial histogram for `title`, 5000 distinct-title target, 20 buckets

3.2 Q2.2 Histogram Construction Time

Column	1000 target	3000 target	5000 target
id	0.424495 s	1.051850 s	2.039610 s
title	0.395126 s	1.181556 s	2.211864 s

Table 2: Average histogram construction time over 3 trials (20 buckets)

For **title**, the average raw rows required to obtain the target number of distinct titles were 1043.67, 3181.33, and 5416.33 for targets 1000, 3000, and 5000, respectively.

3.3 Q2.3 Frequency Bucket Boundaries

The following tables contain the exact boundaries from the latest 20-bucket CSV outputs.

ID – 1000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	5304.0	143879.0	50
2	150684.0	255772.0	50
3	256211.0	366171.0	50
4	366374.0	479795.0	50
5	482919.0	622726.0	50
6	629152.0	784399.0	50
7	785823.0	945461.0	50
8	947517.0	1068169.0	50
9	1074234.0	1209398.0	50
10	1213937.0	1426404.0	50
11	1438339.0	1564467.0	50
12	1566369.0	1683573.0	50
13	1684123.0	1793262.0	50
14	1793990.0	1907933.0	50
15	1908257.0	2042122.0	50
16	2044253.0	2140858.0	50
17	2141370.0	2230480.0	50
18	2232164.0	2330116.0	50
19	2330820.0	2436275.0	50
20	2439999.0	2528065.0	50

Table 3: ID histogram boundaries for the 1000-sample target

ID – 3000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	1342.0	116658.0	150
2	117189.0	224106.0	150
3	225563.0	319387.0	150
4	320450.0	441467.0	150
5	444114.0	601544.0	150
6	602342.0	787774.0	150
7	788030.0	915487.0	150
8	915767.0	1015188.0	150
9	1016215.0	1142799.0	150
10	1146912.0	1276962.0	150
11	1278164.0	1393670.0	150
12	1393873.0	1572921.0	150
13	1573167.0	1717048.0	150
14	1717411.0	1873267.0	150
15	1873791.0	1980819.0	150
16	1981505.0	2103502.0	150
17	2103689.0	2205675.0	150
18	2206013.0	2285512.0	150
19	2286406.0	2402892.0	150
20	2403037.0	2525548.0	150

Table 4: ID histogram boundaries for the 3000-sample target

ID – 5000 samples

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	545.0	125318.0	250
2	126569.0	284501.0	250
3	285712.0	433603.0	250
4	433941.0	579370.0	250
5	582311.0	710102.0	250
6	710457.0	835445.0	250
7	835624.0	963380.0	250
8	963949.0	1096807.0	250
9	1098059.0	1194013.0	250
10	1194195.0	1285227.0	250
11	1285267.0	1435385.0	250
12	1435534.0	1577596.0	250
13	1577759.0	1699420.0	250
14	1700569.0	1829558.0	250
15	1831093.0	1950061.0	250
16	1950110.0	2064500.0	250
17	2064626.0	2180446.0	250
18	2180595.0	2261249.0	250
19	2261611.0	2393833.0	250
20	2394629.0	2525925.0	250

Table 5: ID histogram boundaries for the 5000-sample target

Title – 1000 distinct titles

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	#3.1: The Cavern	#spiritmastertoopenforthe flaminglips	2
2	(#1.1)	(#1.1)	1
3	(#1.101)	(#1.11727)	4
4	(#1.12)	(#1.12)	1
5	(#1.1201)	(#1.318)	17
6	(#1.33)	(#1.34)	2
7	(#1.3447)	(#1.3624)	4
8	(#1.37)	(#1.39)	3
9	(#1.4)	(#1.4)	1
10	(#1.4382)	(#1.4638)	4
11	(#1.5)	(#1.5)	1
12	(#1.504)	(#1.52)	2
13	(#1.59)	(#1.59)	1
14	(#1.6)	(#1.6)	1
15	(#1.7)	(#1.88)	7
16	(#1.9)	(#1.9)	1
17	(#1.90)	(#2.19)	16
18	(#2.2)	(#2.41)	8
19	(#2.5)	(#2.5)	1
20	(#2.7)	Zwart op wit	923

Table 6: Title histogram boundaries for 1000 distinct titles

Title – 3000 distinct titles

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	#4.2: The Secret	(#1.0)	3
2	(#1.10)	(#1.10)	1
3	(#1.10059)	(#1.1643)	41
4	(#1.17)	(#1.17)	1
5	(#1.171)	(#1.179)	4
6	(#1.18)	(#1.18)	1
7	(#1.185)	(#1.3973)	53
8	(#1.4)	(#1.4)	1
9	(#1.40)	(#1.43)	7
10	(#1.44)	(#1.44)	1
11	(#1.4535)	(#1.499)	8
12	(#1.5)	(#1.5)	1
13	(#1.50)	(#1.78)	46
14	(#1.8)	(#1.8)	1
15	(#1.80)	(#18.161)	50
16	(#2.1)	(#2.69)	28
17	(#2.7)	(#2.7)	1
18	(#2.73)	(#3.48)	34
19	(#3.5)	(#3.9)	5
20	(#30.6)	ZooHood	2713

Table 7: Title histogram boundaries for 3000 distinct titles

Title – 5000 distinct titles

Bucket	Lower Boundary	Upper Boundary	Distinct Values
1	'Debbie's Got Reefer'	'Til Death Do Us Part	3
2	(#1.1)	(#1.1)	1
3	(#1.10)	(#1.10)	1
4	(#1.100)	(#1.158)	55
5	(#1.16)	(#1.16)	1
6	(#1.161)	(#1.199)	16
7	(#1.2)	(#1.2)	1
8	(#1.206)	(#1.2984)	32
9	(#1.3)	(#1.3)	1
10	(#1.30)	(#1.30)	1
11	(#1.3026)	(#1.319)	11
12	(#1.32)	(#1.32)	1
13	(#1.321)	(#1.3975)	18
14	(#1.4)	(#1.4)	1
15	(#1.4005)	(#1.490)	29
16	(#1.5)	(#1.5)	1
17	(#1.50)	(#1.598)	27
18	(#1.6)	(#1.6)	1
19	(#1.60)	(#5.10)	233
20	(#5.14)	Zwei neue auf Schloss Einstein	4566

Table 8: Title histogram boundaries for 5000 distinct titles

3.4 Q2.4 Maximum Selectivity Error

Column	1000 distinct	3000 distinct	5000 distinct
id	0.000000	0.000000	0.000000
title	0.001592	0.001316	0.001960

Table 9: Average maximum selectivity error over 3 trials

For `id`, the error is zero because every sampled ID occurs exactly once. For `title`, duplicate values have different frequencies, producing a non-zero estimation error.

3.5 Q2.5 Effect of Sample Size

For `title`, the sample-size target is a **distinct-value target**, not a raw-row target.

Measure	1000 target	3000 target	5000 target
id distinct values	1000	3000	5000
title distinct values	1000	3000	5000
id raw rows (avg)	1000.00	3000.00	5000.00
title raw rows (avg)	1043.67	3181.33	5416.33
id build time (s, avg)	0.424495	1.051850	2.039610
title build time (s, avg)	0.395126	1.181556	2.211864
id objective (avg)	0.000000	0.000000	0.000000
title objective (avg)	13.751662	152.745731	629.932180
id max error (avg)	0.000000	0.000000	0.000000
title max error (avg)	0.001592	0.001316	0.001960

Table 10: Effect of target distinct-value count on the optimal serial histogram

The latest experiment obtained exactly 1000, 3000, and 5000 distinct titles. The average raw rows required were 1043.67, 3181.33, and 5416.33, respectively.

3.6 Q2.6 Optimal Histogram Objective

The optimal serial histogram minimizes

$$\sum_{i=1}^B n_i V_i$$

where n_i is the number of distinct values in bucket i and V_i is the variance of their frequencies.

Column	1000 distinct	3000 distinct	5000 distinct
id	0.000000	0.000000	0.000000
title	13.751662	152.745731	629.932180

Table 11: Average optimal histogram objective values

3.7 Q2.7 Full-table Time Extrapolation

A least-squares linear fit was used:

$$T(n) = \text{slope} \cdot n + \text{intercept}.$$

For `id`, the regression uses raw sampled rows:

$$T_{\text{id}}(n) = 0.000403778708n - 0.039350791783.$$

For the full-table cardinality

$$n = 2,528,312,$$

the extrapolated time is

$$1020.839203 \text{ s} \approx 17.01 \text{ minutes.}$$

For `title`, the regression uses the average raw rows actually processed to obtain the distinct-title targets:

$$T_{\text{title}}(n) = 0.000415821220n - 0.073508272838.$$

The extrapolated full-table time is

$$1051.252271 \text{ s} \approx 17.52 \text{ minutes.}$$

These are extrapolations from three measured points, not direct full-table measurements.

3.8 Q2.8 Sample Size vs. Construction Time

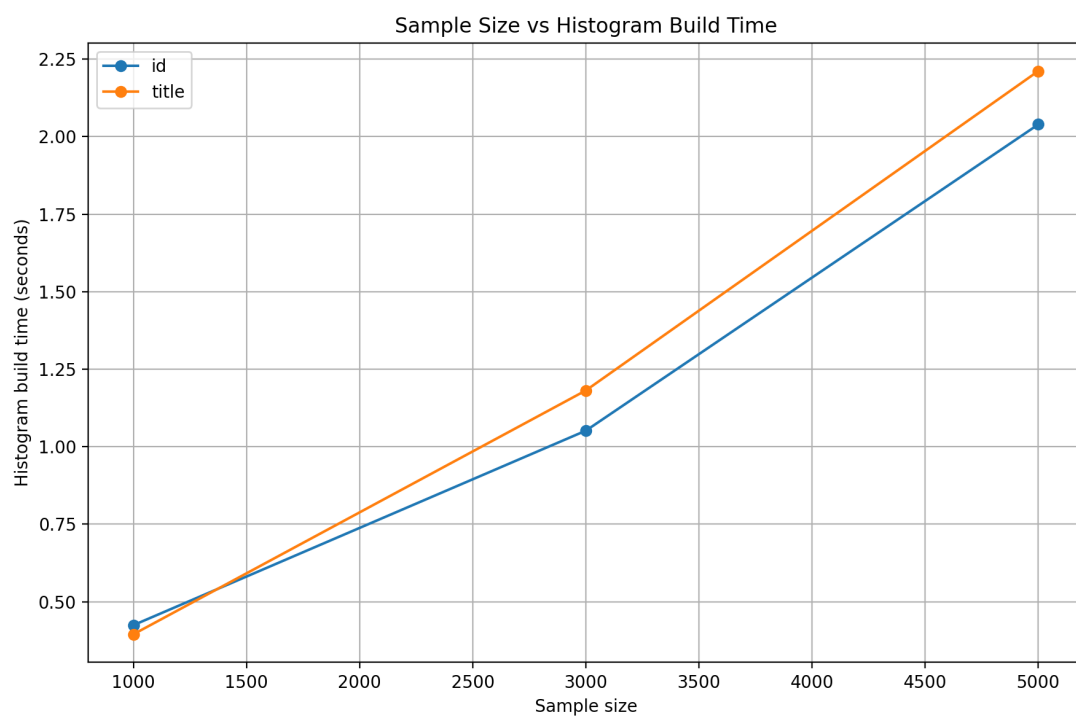


Figure 9: Target size versus average histogram construction time

3.9 Q2.9 Sample Size vs. Maximum Selectivity Error

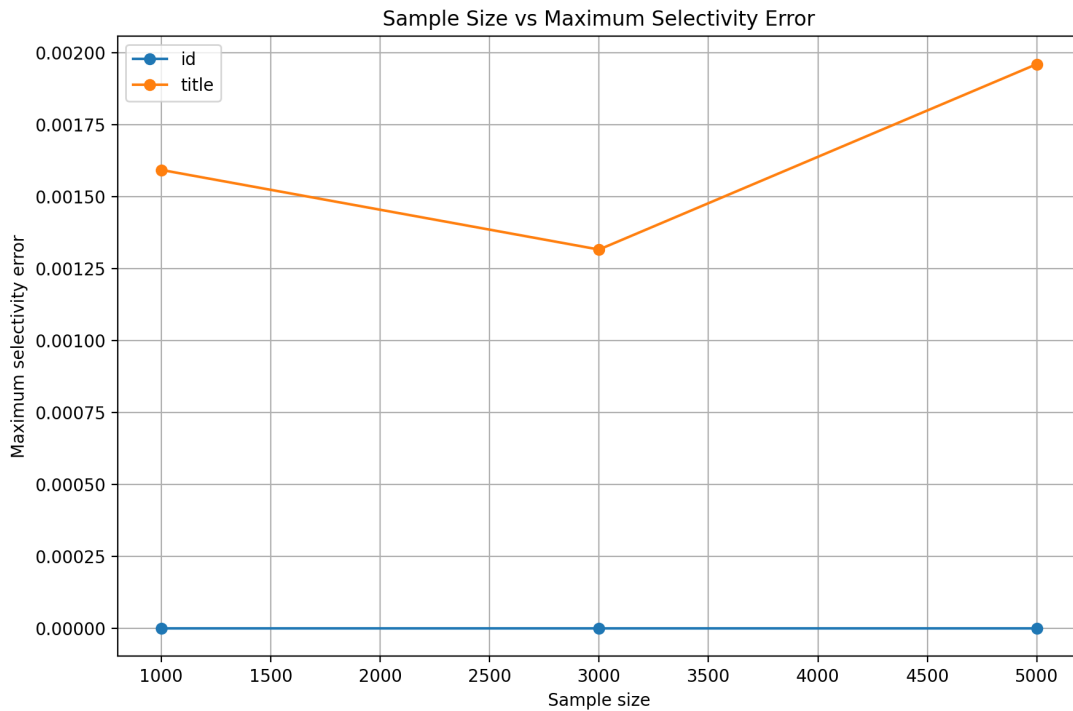


Figure 10: Target distinct-value count versus average maximum selectivity error

3.10 Q2.10 Number of Buckets

The experiment uses 20 buckets, satisfying the assignment’s 10–50 bucket requirement. More buckets generally preserve more information about the frequency distribution and can reduce selectivity estimation error, at the cost of additional boundary information and construction work.

4 Final Results Summary

All values below are averages over 3 independent trials. For `id`, the target is the number of raw sampled rows. For `title`, the target is the number of distinct titles, while `Raw Rows` reports the actual average number of rows required to reach that distinct count.

Column	Target	Raw Rows (avg)	Time (s, avg)	Buckets	Objective (avg)	Max Error (avg)
id	1000	1000.00	0.424495	20	0.000000	0.000000
title	1000 distinct	1043.67	0.395126	20	13.751662	0.001592
id	3000	3000.00	1.051850	20	0.000000	0.000000
title	3000 distinct	3181.33	1.181556	20	152.745731	0.001316
id	5000	5000.00	2.039610	20	0.000000	0.000000
title	5000 distinct	5416.33	2.211864	20	629.932180	0.001960

Table 12: Final Q2 experimental results (3-trial averages, 20 buckets)

5 Final Results Summary

All build-time and maximum-selectivity-error figures below are averaged over 3 independent trials per (column, sample size) combination. This experiment used **20 buckets**.

Column	Samples	Raw Rows (avg)	Time (s, avg)	Buckets	Objective (avg)	Max Error (avg)
id	1000	1000	0.324314	20	0.000000	0.000000
title	1000	1000	0.288884	20	8.571589	0.000993
id	3000	3000	1.024027	20	0.000000	0.000000
title	3000	3000	1.060286	20	117.892466	0.001507
id	5000	5000	1.939624	20	0.000000	0.000000
title	5000	5000	1.964463	20	530.520648	0.002106

Table 13: Final Q2 experimental results (3-trial averages, 20 buckets)

6 Final Results Summary

All build-time and max-error figures below are averaged over 3 independent trials per (column, sample size) combination. This run used **5** buckets.

Column	Samples (target)	Raw Rows (avg)	Time (s, avg)	Objective (avg)	Max Error (avg)
id	1000	1000	0.064031	0.000000	0.000000
title	1000	4501	0.275596	1538.960662	0.004186
id	3000	3000	0.194093	0.000000	0.000000
title	3000	4501	0.290654	1849.632632	0.004871
id	5000	5000	0.367282	0.000000	0.000000
title	5000	7501	0.550376	4499.977598	0.003829

Table 14: Final Q2 experimental results (3-trial averages, 20 buckets)